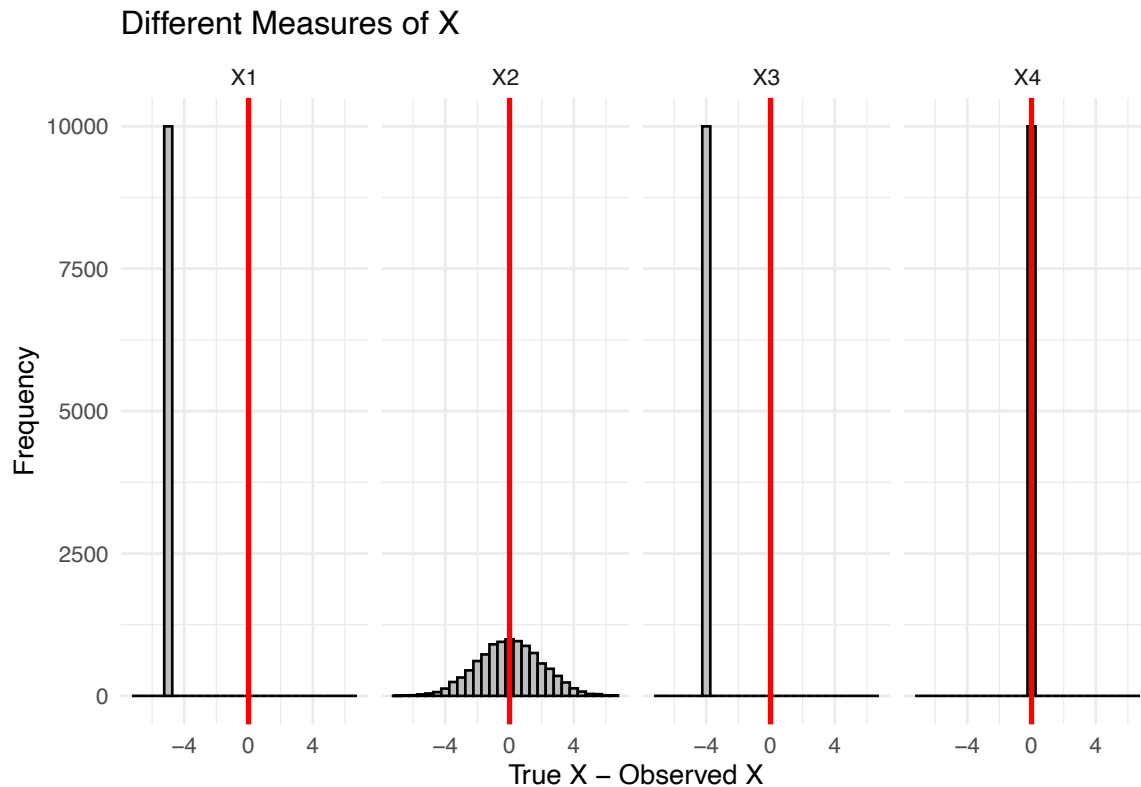


POLI110

AODI

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Question: The following chart shows the performance of four different measures (X_1 , X_2 , X_3 , X_4) of variable X . The x-axis shows the difference between the observed value of X and the true value from each measure ($X_{Observed} - X_{True}$). The y-axis shows how frequently this difference occurs across 10,000 cases. The dark vertical line indicates where $X_{True} - X_{Observed} = 0$ (the measure exactly gets the right value for the variable).



First, we need to clarify what the X and Y axes represent. Here, the X axis represents the values of the random errors, i.e., the differences between each observation and its true value. If an observation equals the true value, then the difference between them should be 0. The Y axis represents the frequency, more specifically, the frequency of occurrence of the different error values. We will now discuss two different scenarios:

-Random error: Random error means that the measured value deviates randomly from the true value during the measurement. When visualising this type of error, we usually expect the errors to be randomly distributed around 0. So, suppose you see error values on the chart randomly spread around the red line (the baseline)

representing 0 error) without any apparent pattern or tendency to cluster. In that case, this is a manifestation of random error.

-Systematic error: A systematic error is a systematic deviation of the measured value from the actual value in a certain direction. A histogram usually shows this by a distinct peak or concentrated distribution outside the red vertical line (i.e., the 0 error line). This means that the error values (the values on the X-axis) are not randomly distributed but concentrated in a certain range, showing a systematic bias.

